

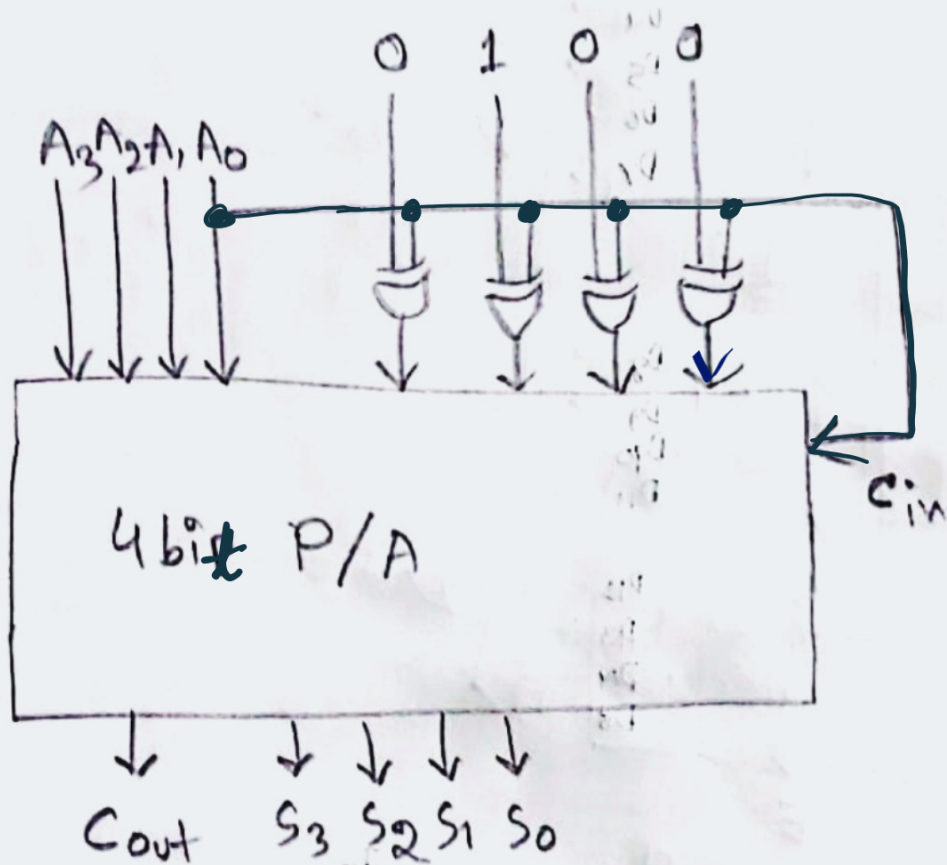
1 $A = A_3 A_2 A_1 A_0$ (MSB LSP)

if A is odd

output = 14 - 4

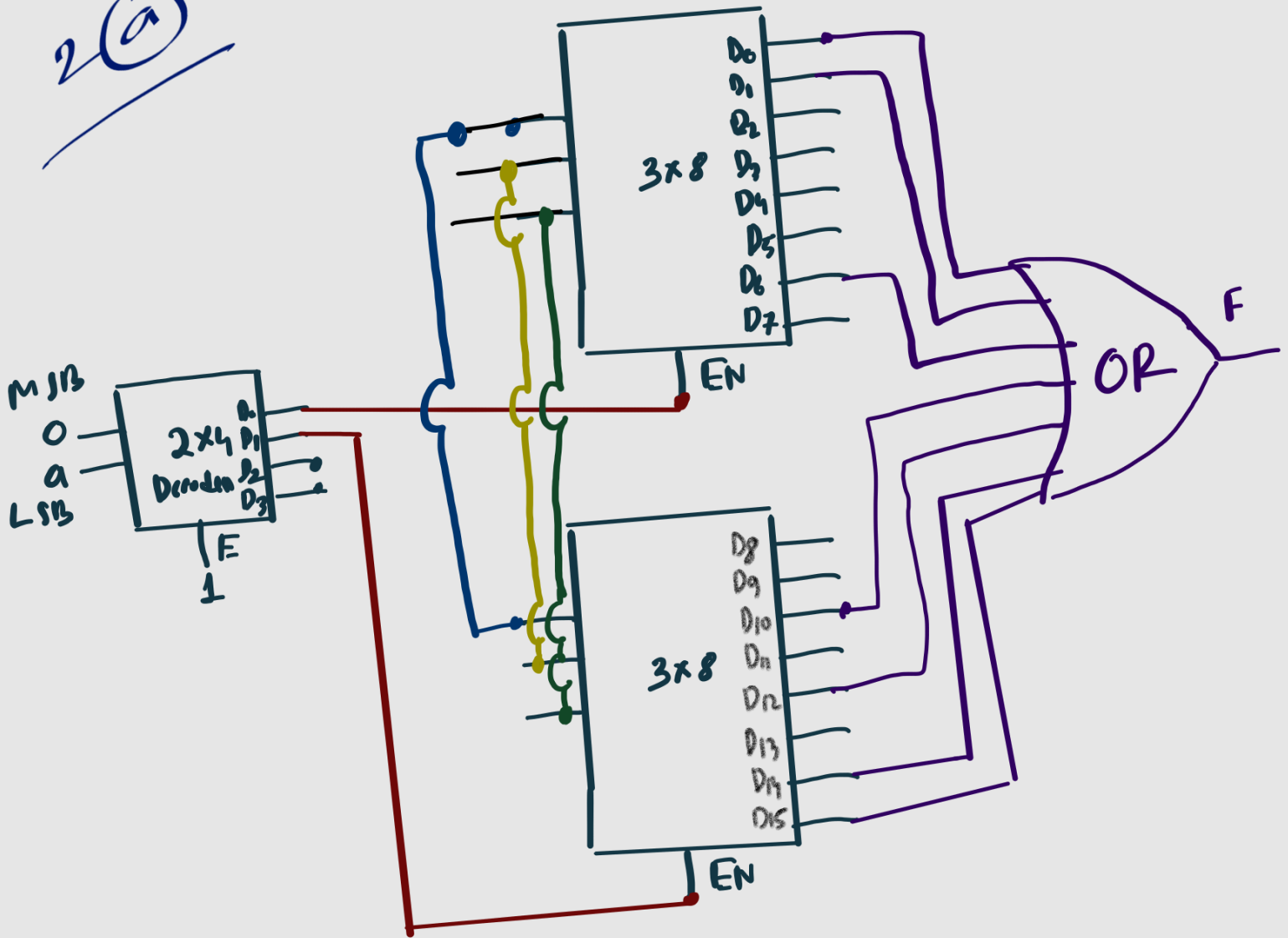
if A is even

output = A + 4.



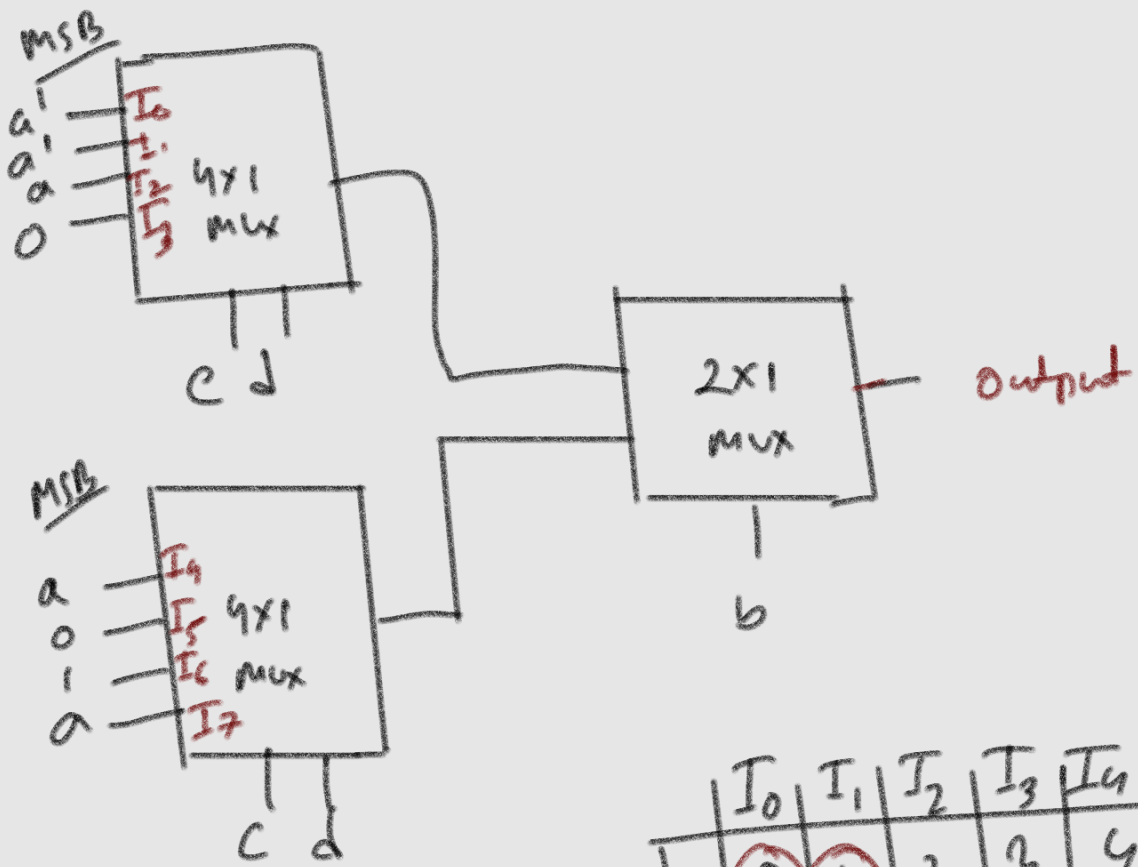
$$F(a, b, c, d) = \sum (0, 1, 6, 10, 12, 14, 15)$$

2(a)



2(b)

$$F(a, b, c, d) = \sum (0, 1, 6, 10, 12, 14, 15)$$

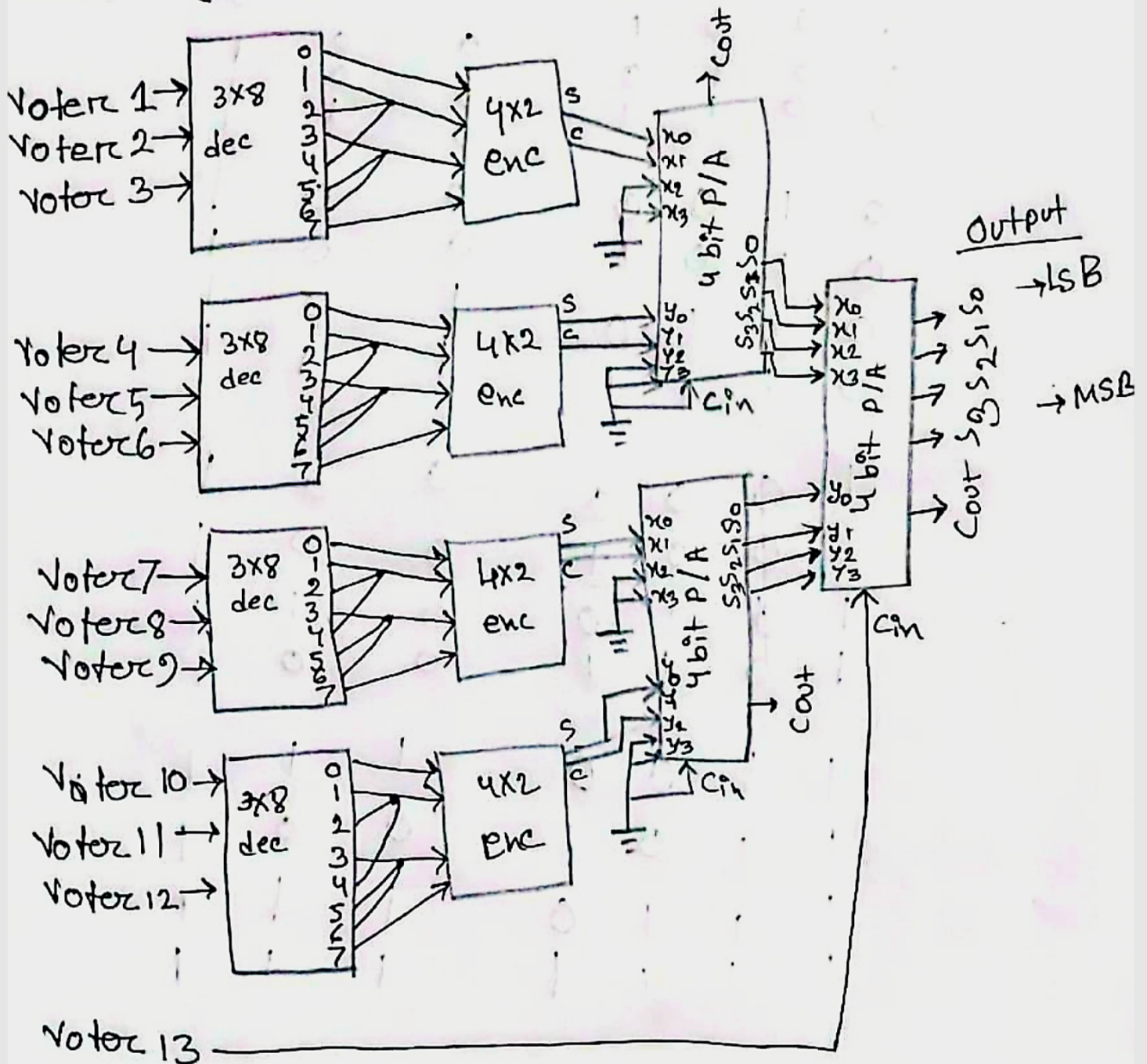


	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
a'	0	1	2	3	4	5	6	7
a	8	9	10	11	12	13	14	15
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	a'	a'	a	0	a	0	1	a

3

3/ 13 person voter counting system

~~2/13~~ (Encoder and decoder used to build full adder)



4

D₃

D₂

D₁

D₀

A

B

0

0

0

0

0

x

x

1

0

0

0

1

0

0

2

0

0

1

0

0

1

3

0

0

1

1

0

1

4

0

1

0

0

1

0

5

0

1

0

1

0

0

6

0

1

1

0

1

0

7

0

1

1

1

1

0

8

1

0

0

0

1

1

9

1

0

0

1

0

0

10

1

0

1

0

0

1

11

1

0

1

1

0

0

12	1	1	0	0	1	1
13	1	1	0	1	0	0
14	1	1	1	0	1	1
15	1	1	1	1	1	1

$$A = \sum (4, 6, 7, 8, 12, 14, 15) + d(0)$$

$$B = \sum (2, 3, 8, 10, 12, 14, 15) + d(0)$$

→ Then draw K-Map to find the function and generate the circuit (i)

